Mathematical Modeling and Simulation of Quantum Gates Operations Using Qiskit Project Report:

1. The project

A quantum computer works by applying small operations, called gates, to quantum bits, called qubits. We took the basic one-qubit gates (X, Y, Z, H and the three rotations Rx, Ry, Rz) and asked a simple question: if we work out by hand what each gate should do, does the Qiskit simulator agree? We tested this on four starting states, a sweep of 73 rotation angles, three gate-order comparisons and a repeated-measurement experiment. The simulator matched the hand calculations to about 15 decimal places, and the repeated-measurement errors shrank exactly as probability theory predicts.

2. The key ideas

Term	Plain meaning
Qubit	A bit that can be 0, 1, or a mixture of both until measured
Superposition	The "mixture": the qubit has an amplitude for 0 and an amplitude for 1
Gate	An operation that changes a qubit's state, written as a 2×2 grid of numbers (a matrix)
Unitary	A gate that never loses or creates probability: the total stays 100% and the gate can be undone
Measurement	Looking at the qubit; it gives 0 or 1 with probabilities set by the amplitudes
Bloch sphere	A globe where every qubit state is an arrow.
Global phase	A hidden overall twist (like multiplying by −1) that no experiment can detect, so two states that differ only by it are physically the same
Fidelity	How alike two states are: 1 means identical, 0 means completely different
Shots	The number of times a circuit is run and measured to estimate probabilities

The states used in the project are $|0\rangle$ (north pole), $|1\rangle$ (south pole), $|+\rangle$ (on the equator, a 50/50 mix of 0 and 1) and $|+i\rangle$ (on the equator, a quarter turn from $|+\rangle$).

3. What each gate does

- **X** flips 0 and 1, like a NOT gate. It sends the north pole to the south pole.
- **Y** also flips 0 and 1, but adds a twist to the phase.

- **Z** leaves 0 alone and flips the sign of 1. It changes the phase, not the probabilities of 0 and 1.
- **H (Hadamard)** turns a definite state into a 50/50 superposition. It moves the north pole onto the equator.
- **Rx, Ry, Rz** rotate the arrow by a chosen angle around the x, y or z axis, so they give a smooth control instead of a fixed flip.

4. What we did and what we found

Experiment 1: are the gates valid?

We checked that each gate is unitary. The largest error was 3×10^{-16}, which is just computer rounding. So every gate is a valid quantum operation.

Experiment 2: four gates on four starting states

We applied X, Y, Z and H to each of the four states (16 cases) and compared Qiskit to the hand calculation. They matched in every case, with fidelity within 9×10^{-16} of 1. The important lesson was that Z looks like it does nothing to $|0\rangle$, but it clearly turns $|+\rangle$ into $|-\rangle$. That is why a single starting state is not enough.

Experiment 3: rotation sweep

We rotated by every 5° from 0° to 360°. On $|0\rangle$, Rx and Ry make the probability of seeing 1 rise and fall smoothly (0 at 0°, 0.5 at 90°, 1 at 180°). Rz does nothing visible to $|0\rangle$, because $|0\rangle$ sits on the rotation axis. But on $|+\rangle$, Rz spins the arrow all the way around the equator, even though the measured probabilities stay at 0.5. All 438 points matched the formulas.

Experiment 4: does order matter?

Applying H then Z gives a different state from Z then H. On $|0\rangle$ they give $|-\rangle$ and $|+\rangle$, which are completely different states (fidelity 0) but give identical 50/50 results if you measure in the usual way. The same is true for X and H. We also confirmed the identity "H, then Z, then H equals X": sandwiching Z between two Hadamards turns a phase flip into a bit flip.

Experiment 5: measuring many times

Measuring $H|0\rangle$ should give 1 half the time. With few shots the estimate is noisy (typical error 0.052 at 100 shots). With 10,000 shots it is 0.005. The error shrinks like 1 divided by the square root of the number of shots, which matches the standard binomial formula. A 100-fold increase in shots gives about a 10-fold drop in error.